

1. Token

Definition

A **token** is the smallest unit of text obtained after breaking a sentence into individual words, numbers, or punctuation marks (depending on the tokenization method). In basic NLP, each word is usually treated as one token.

Example

Sentence:

Natural Language Processing is interesting.

Tokens:

- Natural
- Language
- Processing
- is
- interesting

Total Tokens = 5

2. Type

Definition

A **type** is a **unique token** in a text. Repeated occurrences of the same word are counted only once.

Example

Sentence:

AI is transforming AI applications.

Tokens:

AI, is, transforming, AI, applications

Unique words (Types):

- AI
- is
- transforming
- applications

Total Tokens = 5

Total Types = 4

3. Type-Token Ratio (TTR)

Definition

The **Type-Token Ratio (TTR)** is a statistical measure used to evaluate the **lexical diversity** of a text. It represents the proportion of unique words (types) to the total number of words (tokens).

Formula: $TTR = \text{No of Types} / \text{No of Tokens}$

Example

Types = 20

Tokens = 25

$$TTR = 20/25 = 0.80$$

4. Interpretation of TTR

The TTR value indicates the lexical diversity of a text.

TTR Value	Interpretation
Greater than 0.70	High lexical diversity (more unique words, fewer repetitions)
0.50 – 0.70	Moderate lexical diversity
Less than 0.50	Low lexical diversity (more repeated words, limited vocabulary)

Example 1

Tokens = 40

Types = 36

$$TTR = 36/40 = 0.90$$

Interpretation: The passage has **high lexical diversity**, indicating that it contains many unique words and very few repeated words.

Example 2

Tokens = 40

Types = 18

$$TTR = 18/40 = 0.45$$

Interpretation: The passage has **low lexical diversity**, indicating that many words are repeated and the vocabulary is less varied.

5. Relative Frequency (Probability of Occurrence)

Definition

The **probability of occurrence** of a word is the likelihood of that word appearing in the given text. It is calculated by dividing the frequency of the word by the total number of tokens.

Formula

$P(w) = \text{Frequency of the Word} / \text{Total Number of Tokens}$

Example

Word "**customer**" appears **4** times.

Total tokens = **40**

$P(\text{customer}) = 4/40 = 0.10$

Examples:

1. Smart traffic systems are helping cities reduce congestion. Many vehicles now use GPS navigation to find faster routes. Traffic signals collect data from sensors to manage vehicle flow efficiently. Smart transportation improves road safety and reduces travel time for commuters.

Solution:

a) **Total Tokens = 40**

Total Types = 37

TTR = 37/40 = 0.925

Interpretation:

Since **TTR > 0.70**, the passage has **high lexical diversity**.

b) Word = **traffic**

Frequency = **2**

$P(\text{traffic}) = 2/40 = 0.05$

2. स्मार्ट कृषि तकनीक किसानों की मदद कर रही है। कई किसान मिट्टी की नमी मापने के लिए सेंसर का उपयोग करते हैं। सेंसर से प्राप्त जानकारी के आधार पर सिंचाई की जाती है। इससे पानी की बचत होती है और फसल की गुणवत्ता में सुधार होता है।

Solution:

a) **Total Tokens = 47**

Total Types = 39

TTR = 39/47 = 0.83

Interpretation:

Since **TTR > 0.70**, the passage has **high lexical diversity**.

b) Word = **सेंसर**

Frequency = **2**

$P(\text{सेंसर}) = 2/47 = 0.043$